

FIPS-Aligned STARK Verification of ML-DSA and Classical Signatures, with an AIR-Based Cost Model

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Abstract. *Completing a FIPS-aligned STARK framework into a deployable signature-verification system*, we present STARK proofs of full signature verification at NIST post-quantum security Levels 1, 3, and 5 for four schemes — RSA-2048, ECDSA-P256, Ed25519, and the post-quantum FIPS-204 ML-DSA standard — end-to-end under native SHA-3, with concrete QROM soundness (knowledge soundness in the classical ROM); because the STARK is itself post-quantum, this delivers post-quantum verification even for the classical signatures. Building on a DEEP-ALI merge at folding arity $k=4$, we construct *witness-binding* AIRs for RSA-2048, ECDSA-P256 and Ed25519 (the latter two at full $K=256$) and a four-sub-AIR ML-DSA composite, so a *tampered signature is rejected in-circuit*, not merely shown low-degree. Binding the witness makes the sound proof *trace-opening-dominated*: 9.8/35/165 MiB at L1 for RSA-2048/Ed25519/ECDSA-P256 and 17–55 MiB for the ML-DSA composite — the $2rw$ -cell openings dwarfing a sub-MiB, width-independent LDT (0.40–2.2 MiB) — verifying in 25–930 ms under a single FIPS-202 assumption.

A *cost model* parameterizes an AIR’s NIST Level 1/3/5 STARK cost as two components: the merge collapses the width- w trace into a single composition polynomial for low-degree testing over $|H_0|=Tb$, making the *LDT component* a parameter-free, width-independent closed form, while witness-binding reopens the width- w trace rows at each query, adding an $O(rw)$ trace-opening term. Consequently, the sound proof size and verification time are signature-specific and binding-dominated (Ed25519 proves $\approx 6.7\times$ slower than RSA-2048 despite a shorter trace; ECDSA-P256 verifies in 320 ms vs RSA’s 25). Under both FRI ($\div 4$, a constant-rate 4:1 fold per round) and STIR ($\div 2$, the rate-halving schedule that folds 4:1 but improves the rate each round) the $1.15\text{--}1.41\times$ LDT-component edge is *second-order* at whole-proof, so we deploy the unconditionally sound $\div 4$ for simplicity. The result is a FIPS-only, transparent, post-quantum-verifiable path for all four signature schemes, with an open-source artifact.

Keywords: Post-quantum cryptography · Zero-knowledge proofs · STARKs · STIR · SHA-3 · FIPS compliance · QROM

1 Introduction

Scalable Transparent Arguments of Knowledge (STARKs) [1] require no trusted setup; their post-quantum security rests on two pillars, information-theoretic soundness of a low-degree test and collision resistance of a Merkle hash. In practice production systems compromise one: Plonky2 [12], Plonky3 [13], and STWO [10] place an algebraic hash (Poseidon, Rescue Prime) on the Merkle path, capping security at its (often unstated ~ 80 -bit) margin with no mature PQ analysis [8,9] and no FIPS 140-3 certification; Winterfell [14] offers SHA3-256 but retains the FRI topology where each layer opens $\Theta(m)$ elements, inflating proofs enough that its developers default to the non-FIPS BLAKE3. Neither path is simultaneously compact, QROM-secure, and FIPS-aligned.

1.1 Relationship to the companion [19]

This paper builds on the FIPS-aligned DEEP-ALI merge of the companion [19], inheriting *unchanged* its QROM decomposition, binding-wall theorem, FIPS-202 capacity argument, and Poseidon-erasure lemma (restated as propositions in Appendices A–B, so this paper’s novelty and soundness are assessable standalone; full proofs in [19], in the ESORICS 2026 proceedings), with HVZK and coset-proximity restated for higher arity. Three things change: (i) the merge submits the ALI-*quotient* $Q = \Phi/Z_H$ ($d_0 = (d_c - 1)T - 1$), which at $d_c = 2$ halves $|H_0|$; (ii) the instantiation uses $d_c = 2$, $|H_0| = 2^{27}$ (vs $d_c = 4$, 2^{29} in [19, Cor. 1]); and (iii) STIR [17] replaces FRI as the LDT, so proof-size comparisons are FRI-vs-STIR within this paper’s single harness (§8). Substituting STIR for FRI under STIR’s unconditional Johnson-regime proximity-gap bound ($\frac{1}{2} \log_2(1/\rho_0) \approx 2.5$ bits/query at $\rho_0 = 1/32$, matching FRI under BCIKS [4, Thm. 8.3]) gives: a per-round shrinkage schedule $t_i = \lceil (\lambda + \log_2(M+2) + 1) / (-\log_2(\sqrt{\rho_i} + \eta_i)) \rceil$ (Theorem 4) yielding $\sum t_i \in \{279, 413, 544\}$ at L1/L3/L5 vs FRI’s $r \cdot M \in \{440, 648, 864\}$ ($\approx 37\%$ analytic query reduction, with the non-deployability caveat of §8.3); a round-count drop $L=16 \rightarrow M=8$ at $k=4$ that lets native SHA-3 displace Poseidon on the prover hot path; a DFMS FS-penalty reduction $2 \log_2(L+2) \rightarrow 2 \log_2(M+2)$ raising the L5 binding wall by one bit; and a cross-LDT empirical comparison on real cryptographic AIRs. The sole computational assumption remains collision resistance of a single FIPS-202 hash. STIR’s stronger *capacity-regime* proximity-gap ($\log_2(1/\rho_0) \approx 5$ bits/query) is conjectural beyond small arities; we refuse it and present only unconditional Johnson-regime parameters.

Contributions: what this paper adds beyond [19]. The companion established the FIPS-aligned DEEP-ALI framework and its QROM accounting but did *not* instantiate full cryptographic signature AIRs, benchmark witness-binding end-to-end verification, evaluate STIR under unconditional bounds, or give an AIR-keyed cost model validated on real circuits. This paper supplies all four as *three* standalone contributions, bridging provable QROM accounting with full-system cryptographic-AIR deployment. (i) *FIPS-aligned, witness-binding STARK proofs of signature verification at NIST post-quantum security Levels 1/3/5* for

four full signature AIRs (RSA-2048, ECDSA-P256, Ed25519, and the FIPS-204 ML-DSA composite), so a *tampered signature is rejected in-circuit*. (ii) A *two-component cost model* (§8.4): the merge makes the LDT width-independent, while witness-binding adds an $O(rw)$ trace-opening term that *dominates* the sound proof (a Python calculator predicts it for any AIR). (iii) The finding that STIR’s rate-improving schedule is *second-order* ($\approx 1\%$) at whole-proof. What is *not* new we state plainly: the LDT substitution is close to mechanical, and the QROM decomposition, binding wall, FIPS-202 limit and HVZK/AoK/Poseidon-erasure are inherited from [19] (Appendix A).

(1) *FIPS-aligned, witness-binding signature verification*. End-to-end verification of four full signature AIRs — an exp-chain RSA-2048, a multi-row ECDSA-P256 and a verdict-enforced Ed25519 (the latter two at full $K=256$), and the FIPS-204 ML-DSA-44/65/87 composite — all under native SHA-3 with concrete QROM bounds at NIST L1/3/5 (Table 5). The trace is Merkle-committed and re-checked at every LDT query, so a *tampered signature is rejected in-circuit*, not merely shown low-degree; the ML-DSA composite binds its four sub-AIRs’ shared cells via three composed mechanisms — F2b TraceOpening Merkle inclusion, merge-layer OOD evaluation at $z \in \mathbb{F}_{p^e} \setminus H_0$ (Schwartz–Zippel slack $\leq 2^{-347}/2^{-475}$ at $\mathbb{F}_{p^6}/\mathbb{F}_{p^8}$), and PI-hash binding (Lemma 1). RSA-2048, Ed25519, ECDSA-P256 are *classical*; only ML-DSA is a NIST PQ standard — the post-quantum guarantee is on the *STARK*, keeping verification transparent and SHA-3-only. To our knowledge no prior STARK verifies FIPS-204 ML-DSA with FIPS-only commitments and concrete multi-level QROM bounds.

(2) *Two-component, AIR-keyed cost model*. The merge collapses the width- w trace into one composition polynomial, so the LDT proof size is a *parameter-free, width-independent* closed form in $(|H_0|, s)$; witness-binding then adds an $O(rw)$ trace-opening term that *dominates* the sound proof, making the total signature-specific. A Python calculator [20] predicts S_{sound} for any AIR from its (T, b, w, r) , self-validated against the measured proofs to $< 1\%$ (§8.4).

(3) *Matched-arity LDT comparison with both schedules sound*. At matched arity $k=4$ we compare STIR’s rate-improving $\div 2$ fold against a *sound* FRI- $k=4$ baseline in the same merge family: both query *every* round (Lemma 3), differing only in schedule. The $1.15\text{--}1.41\times$ LDT-byte advantage at the unconditional Johnson rate (Table 4) is an *LDT-component* ratio — *second-order* ($\approx 1\text{--}2\%$) on the MiB-scale sound proofs, which the $2rw$ openings dominate (§8.3). We deploy the constant-rate $\div 4$ uniform- r fold throughout (structurally FRI- $k=4$): the LDT choice is second-order, as the cost model predicts.

(4)–(6) *LDT-substitution mechanism, plain QROM soundness, inherited framework*. The STIR substitution drops the deployed Merkle workload $\approx 2.1\times (r(L+1) \rightarrow r \cdot M)$ and trims the DFMS FS penalty to ≈ 6.6 bits ($K=10$ vs 18), letting native SHA-3 hold the prover hot path. Plain QROM soundness is Theorem 4 $\circ$ Theorem 1, re-instantiating [19, Theorem 4] with ε_{IT} updated to STIR’s contraction (knowledge soundness not claimed, open for STIR as for FRI before [7]). The merge construction, binding wall and FIPS-202 limit ([19, §5.2–

5.3]), HVZK (Lemma 2), AoK (Theorem 3 via [19, Thm. 9]), and Poseidon-erasure ([19, Lem. 3]) are inherited and restated in Appendix A.

Target regime. Single-signature proofs are multi-MiB and minutes for the large-circuit composites (55.1 MiB / 34 min at the extreme); the intended regime is *amortized*, where a recursive outer STARK aggregates many inner proofs — 789 KiB per signature at $N=4$ (§8.5) — so “deployed” throughout means a pre-computed proof verified once, not single-shot on-line proving.

2 Preliminaries

Goldilocks field. $\mathbb{F}_p$ with $p = 2^{64} - 2^{32} + 1$, chosen for efficient 64-bit arithmetic and high two-adicity ($2^{32} \mid p-1$).

Extension towers. For extension degree e , $|\mathbb{F}_{p^e}| = p^e \approx 2^{64e}$. We construct: $\mathbb{F}_{p^3} = \mathbb{F}_p[x]/(x^3 - x - 1)$; $\mathbb{F}_{p^6} = \mathbb{F}_{p^3}[y]/(y^2 - \alpha)$ for non-square $\alpha \in \mathbb{F}_{p^3}$; $\mathbb{F}_{p^8} = \mathbb{F}_{p^2}[y]/(y^4 - \beta)$ over $\mathbb{F}_{p^2} = \mathbb{F}_p[x]/(x^2 - 7)$ for non-square $\beta \in \mathbb{F}_{p^2}$ not of the form $-4c^4$.¹ Challenge spaces have cardinality 2^{192} , 2^{384} , and 2^{512} respectively. The choice of e controls the Fiat–Shamir ceiling relative to the target security parameter λ ; §5 shows $\mathbb{F}_{p^3}$ is insufficient for L1 at practically relevant q , $\mathbb{F}_{p^6}$ suffices for L1/L3, and $\mathbb{F}_{p^8}$ is needed for L5.

Reed–Solomon codes and the target-degree convention. For trace length T , max constraint degree d_c , trace columns interpolated on $H = \langle \omega_T \rangle$, the α -combined constraint polynomial has $\deg \Phi \leq D := d_c(T - 1)$ and the DEEP-ALL-merge target degree is the quotient bound $d_0 := (d_c - 1)T - 1$ for Φ/Z_H ($Z_H = X^T - 1$; eq. (2)). This is the one place the target differs from [19], which tests at the un-quotiented composition bound $d_c T - 1$ under its convention $D_{\text{pred}} = d_c T$; since the merge submits the quotient $Q = \Phi/Z_H$, $(d_c - 1)T - 1$ is the tight target, halving $|H_0|$ at $d_c=2$. $\text{RS}_{H_0}(d_0)$ is the code of degree- $\leq d_0$ polynomials on the disjoint coset $H_0 = \gamma G \subset \mathbb{F}_p$ ($G \leq \mathbb{F}_p^\times$ smooth of order $|H_0|$, $\gamma \notin G$, §3.2); γ is the coset generator, distinct from the Johnson offset η_i^J . $|H_0|$ is a power of two with $|H_0| = d_0/\rho_0$ at target rate $\rho_0 \approx 1/32$. The deployed L1 instance sets $T = 2^{22}$, $d_c = 2$ ($|H_0| = 2^{27}$); benchmarks (Section 8) use Fibonacci with $T = 2^{16}$, $d_c = 2$ ($|H_0| = 2^{21}$). For functions $f, g : S \rightarrow \mathbb{F}$, $\Delta(f, g) = |\{s : f(s) \neq g(s)\}|/|S|$; $\text{dist}(f, \mathcal{C}) = \min_{g \in \mathcal{C}} \Delta(f, g)$.

STIR as the low-degree test. STIR [17] is a proximity test for $\text{RS}_{H_0}(d_0)$ in M rounds; each round reduces the target degree by folding factor k and halves the domain ($|H_{i+1}| = |H_i|/2$, $d_{i+1} = d_i/k$), so $\rho_{i+1} = (2/k)\rho_i$ improves by $k/2$. We take $k = 4$, $M = 8$, $K = M + 2 = 10$, giving $\rho_i = \rho_0/2^i$ and terminal $|H_M| = 2^{19}$ at $|H_0| = 2^{27}$. We adopt the *unconditional Johnson-regime* proximity gap: $\sigma_{\text{STIR}} \geq 1 - \sqrt{\rho_0} - \text{negl}$, i.e. $\frac{1}{2} \log_2(1/\rho_0) = 2.5$ bits/query at $\rho_0 = 1/32$ (the $\eta=0$ idealization; the deployed η -included yield is $-\log_2(\sqrt{\rho_0} + \eta_0) = 2.43$), identical

¹ Irreducibility of the $\mathbb{F}_{p^8}$ minimal polynomial follows from β non-square (ruling out $y^4 - \gamma^2$) and $\beta \neq -4c^4$ (ruling out the biquadratic split $y^4 + 4c^4 = (y^2 - 2cy + 2c^2)(y^2 + 2cy + 2c^2)$) by the Lang–Niederreiter criterion; $x^2 - 7$ is irreducible since $p \equiv 6 \pmod{7}$.

to FRI under BCIKS [4]. The stronger *capacity-regime* bound (5 bits/query) [17] is conjectural beyond small arities; we do not invoke it, so all soundness holds unconditionally (a future capacity tightening would only shrink t_0). The $|H_i|/2$ halving is the proven mechanism by which the per-round count t_i shrinks across rounds (Theorem 4).

SHA-3 family and sponge capacity. FIPS 202 [15] specifies hash functions built on the Keccak-1600 permutation ($b = 1600$ bits) with capacity $c = b - r_{\text{rate}}$: SHA3-256 has $c = 512$; SHA3-384 has $c = 768$; SHA3-512 has $c = 1024$; SHAKE256 has $c = 512$. The *effective collision entropy* is $n = \min(n_{\text{out}}, c)$, where n_{out} is the output length. For the fixed-output variants $n_{\text{out}} \leq c$, so $n = n_{\text{out}}$. This distinction becomes critical in Section 6.

Collision resistance. Throughout, λ denotes the target security parameter, $\lambda \in \{128, 192, 256\}$ for NIST *post-quantum* security Levels 1, 3, and 5 — the NIST PQC security-strength categories requiring quantum hardness at least that of an AES-128/192/256 key search; later “NIST Level ℓ ” references denote these categories. A hash family $\mathcal{H}$ is $(\varepsilon_{\text{col}}, t)$ -collision resistant if no t -time adversary finds a collision with probability exceeding ε_{col} ; in the post-quantum setting the adversary makes q quantum queries and we write $\varepsilon_{\text{col}}^{\text{pq}}$ for the corresponding bound. In the QROM, $\varepsilon_{\text{col}}^{\text{pq}} \leq c_b q^3 / 2^n$, where $c_b \geq 1$ is the absolute constant from Zhandry’s compressed-oracle bound [5]; we set $c_b = 1$ throughout (a larger c_b shifts κ_{bind} by only $-\log_2 c_b$, negligible here). The IT component depends only on STIR, the computational component only on the instantiated $H \in \mathcal{H}$, and each must remain below $2^{-\lambda}$ at the chosen level.

3 The DEEP-ALI Merge under STIR

3.1 Construction

The merge construction follows [19, §3], with the single change that the target degree is the ALI-quotient bound $d_0 = (d_c - 1)T - 1$ rather than the composition bound $D - 1$ used there (§2). The prover co-commits the trace LDE $T(X)$ (width w , $\deg T \leq T - 1$ per column), the ALI quotient $Q(X) = \sum_{j=1}^s \alpha_j \Phi_j(X) / Z_H(X)$, and a uniform blinder $R(X)$, $\deg R \leq d_0$, into one layer-0 Merkle leaf $(T_1(x), \dots, T_w(x), Q(x), R(x))$. With FS-derived merge-layer DEEP point $z \in \mathbb{F}_{p^e} \setminus H_0$, shift set $\Sigma = \{1, g, \dots, g^{k-1}\}$, vanishing polynomial $V_{z\Sigma}(X) = \prod_{\sigma} (X - \sigma z)$, OOD-interpolant $\tilde{I}$ with $\tilde{I}(\sigma z) = \hat{T}(\sigma z)$, and FS-random $\gamma_1, \gamma_2, \beta$, the verifier checks the ALI identity

$$\sum_{j=1}^s \alpha_j \Phi_j(z) = \hat{Q}(z) \cdot Z_H(z), \quad (1)$$

and the STIR starting function is the DEEP-batched merge

$$C(X) = \gamma_1 X^{d_0 - (T-1-k)} \frac{T(X) - \tilde{I}(X)}{V_{z\Sigma}(X)} + \gamma_2 \frac{Q(X) - \hat{Q}(z)}{X - z} + \beta R(X), \quad (2)$$

with $f_0 = C|_{H_0}$ the STIR starting function and $X^{d_0 - (T-1-k)}$ the ethSTARK degree-correction factor [16]. The layer-0 leaf packs (T, Q, R) into one authentication path per query, independent of trace width or constraint count.

STIR round structure. At round i the transcript supplies a folding challenge α_i and shift challenge $r_i \in \mathbb{F}_{p^e} \setminus H_i$; the prover commits $f_{i+1} = \text{Fold}_{k,\alpha_i}(f_i) - Q_{r_i}$ under root_{i+1} (Q_{r_i} the degree- $(k-1)$ correction onto H_{i+1}). Each query $x \in H_i$ opens $(f_i(x), \text{CP}_i(x))$ under root_i and $f_{i+1}(x^k)$ under root_{i+1} , where CP_i packages the fold residual and the DEEP shift quotient into one leaf, mirroring the FRI-variant merge.

3.2 Soundness of the Merge

$H_0 = \gamma G$ is a coset disjoint from H (by [19, §3.2]; the disjointness is preserved across rounds under $\div 2$ halving because γ generates an odd-order component of $\mathbb{F}_p^\times$ retained by γ^{2^t}). Trace columns are the degree- $\leq (T-1)$ interpolants on H .

The DEEP-ALI degree-bound and initial-distance result (inherited from [19, Theorem 1], restated as Proposition 4 in Appendix A) gives, for an invalid AIR, $\text{dist}(f_0, \text{RS}_{H_0}(d_0)) \geq 1 - \rho_0 - \varepsilon_{\text{ali}}$ with $\varepsilon_{\text{ali}} = (s+T+1)/(|\mathbb{F}_{p^e}| - |H_0|)$ (after $\Phi = Q Z_H + R$ the remainder has $\deg R \leq T-1$, so the OOD check evades only with probability $\leq T/(|\mathbb{F}_{p^e}| - |H_0|)$). At the deployed L1 field ($s = 2^{10}$, $T = 2^{22}$, $d_c = 2$, $|H_0| = 2^{27}$, $|\mathbb{F}_{p^6}| \approx 2^{384}$) this is $\varepsilon_{\text{ali}} < 2^{-361}$ and $\text{dist}(f_0, \text{RS}_{H_0}(d_0)) > 0.968$, exceeding the Johnson threshold $1 - \sqrt{\rho_0} \approx 0.823$, so Theorem 4 applies with margin (the floor still passes at the stressed $|\mathbb{F}_{p^3}| \approx 2^{192}$, $\varepsilon_{\text{ali}} < 2^{-168}$).

3.3 STIR Instantiation and Constant-Path Soundness

Substituting STIR for FRI changes only the low-degree-test layer. Both fold the codeword 4:1 per round ($k=4$); FRI- $k=4$ (“ $\div 4$ ”) holds the rate constant at ρ_0 , whereas under STIR’s “ $\div 2$ ” schedule the rate *halves* each round, $\rho_i = \rho_0(2/k)^i$, and the unconditional Johnson-regime proximity gap gives a *declining* per-round query schedule $t_i = \lceil (\lambda + \log_2(M+2) + 1)/(-\log_2(\sqrt{\rho_i} + \eta_i^J)) \rceil$ (Theorem 4, proof in Appendix B): at $\rho_0=1/32, k=4, M=8, \lambda=128$ this is $\{t_i\} = \{55, 46, 39, 34, 30, 27, 25, 23\}$, $\sum t_i = 279$ (L1; 413/544 at L3/L5). Both LDTs run the same layer-0 merge and open *one* Merkle path per query per round (the k -coset leaf $(\{f_i(\zeta x_i)\}, \text{CP}_i(x_i))$ plus an OOD reply, Lemma 3), querying *every* round (the per-round IT term of Theorem 1 forbids omission); they differ only in the schedule, so $\text{Opens}_{\text{STIR}} = \sum_i t_i < t_{\text{FRI}} M = \text{Opens}_{\text{FRI}}$ (Theorem 5), the ratio growing with ρ_0 as in [17, Table 2]. The per-leaf cost grows with k (the leaf holds k values), offsetting the path-count reduction (§8.1). Public parameters are absorbed into Fiat–Shamir before root_0 , closing a DFMS [6] requirement and separating levels. Statements and proofs (contraction, single-leaf, distance, public-parameter binding) are in Appendix B.

3.4 Cross-Sub-AIR Binding

Lemma 1 (Multi-mechanism cross-sub-AIR binding). *Let $\mathcal{A}_1, \dots, \mathcal{A}_S$ be the witness-sharing sub-AIRs ($S = 4$ for ML-DSA: V17, INTT, COEFF,*

TRANSCRIPT — the deployed composite instantiates these as 9 FRI sub-proofs, with *INTT* run K times and *COEFF* split into three) sharing N witness cells $\{w_n\}$. If each shared w_n is bound by at least one of (a) F2b TraceOpening at a known layer-0 row, (b) merge-layer OOD — the claim $\widehat{T}_n(z)$ at FS-derived $z \in \mathbb{F}_{p^e} \setminus H_0$ agreeing across the sub-AIRs' eq. (1), or (c) PI-hash absorption before any sub-AIR challenge, then a q -bounded quantum prover satisfies all S proofs with inconsistent shared values only with probability $\varepsilon_{\text{cross-bind}} \leq S\varepsilon_{\text{F2b}} + N\varepsilon_{\text{OOD}} + \varepsilon_{\text{PI}}$, where $\varepsilon_{\text{F2b}}, \varepsilon_{\text{PI}} \leq c_b q^3 / 2^n$ (QROM collision resistance of the level's SHA-3 variant [5]) and $\varepsilon_{\text{OOD}} \leq D_n / (|\mathbb{F}_{p^e}| - |H_0|)$ per cell (Schwartz–Zippel on the degree- D_n difference, $D_n = (d_c - 1)T - 1$). At L1 ($N \approx 2^{14}$, $D_n \approx 2^{22}$, $|\mathbb{F}_{p^6}| \approx 2^{384}$, $q \leq 2^{41}$) the aggregate OOD term $N\varepsilon_{\text{OOD}} \leq 2^{-347}$ and $\varepsilon_{\text{cross-bind}} \leq 2^{-128+o(1)}$, dominated by the binding-wall hash term (§6).

Proof. Union bound over the three independent rejection paths; full proof in the full version. $\square$

Since $\varepsilon_{\text{cross-bind}}$ shares the binding-wall hash term, it folds into $\varepsilon_{\text{bind}}$ rather than adding a ceiling below κ_{sys} : at the deployed $q \leq 2^{41}$ it sits near the L1 floor but does not erode κ_{sys} beyond the binding wall already accounted in Table 1 (a stronger adversary erodes both together).

4 Modular Security Decomposition

Theorem 1 (Modular security decomposition). *Let Π be the DEEP-ALL-STIR STARK over $\mathbb{F}_{p^e}$ compiled via Fiat–Shamir with hash H having effective collision entropy $n = \min(n_{\text{out}}, c)$ bits, under STIR's domain-halving schedule with per-round queries $t_i = \lceil (\lambda + \log_2(M+2) + 1) / (-\log_2(\sqrt{\rho_i} + \eta_i)) \rceil$ for $i = 0, \dots, M-1$ at round-rate $\rho_i = \rho_0 \cdot (2/k)^i$. For any (q, t) -QROM adversary $\mathcal{A}$:*

$$\begin{aligned} \varepsilon_{\Pi}(q) \leq & \underbrace{\sum_{i=0}^{M-1} (\sqrt{\rho_i} + \eta_i)^{t_i} + \frac{s+T+1}{|\mathbb{F}_{p^e}| - |H_0|} + \frac{\sum_{i=0}^{M-1} t_i}{|\mathbb{F}_{p^e}| - |H_0|}}_{\varepsilon_{\text{IT}}} \\ & + \underbrace{(M+2) c_b \frac{q^3}{2^n}}_{\varepsilon_{\text{bind}}} + \underbrace{\frac{K^2(q+1)^2}{|\mathbb{F}_{p^e}|}}_{\varepsilon_{\text{FS}}} \end{aligned} \quad (3)$$

where $\sqrt{\rho_i} + \eta_i \leq 1 - \sigma_i$ with $\sigma_i \geq 1 - \sqrt{\rho_i} - \eta_i$ by Theorem 4 (Johnson regime, unconditional), and $K = M+2$. Each t_i satisfies $(\sqrt{\rho_i} + \eta_i)^{t_i} \leq 2^{-(\lambda + \log_2(M+2) + 1)}$, so the M round-by-round contributions sum to at most $2^{-\lambda}$.

Proof. Inherited from the five-hybrid argument of [19, Theorem 4] (full version), restated as Proposition 1 (Appendix A); the STIR substitution changes only ε_{IT} , which absorbs Theorem 4. $\square$

4.1 The Three Ceilings

In bits ($\kappa = -\log_2 \varepsilon$):

$$\begin{aligned} \kappa_{\text{IT}}(\{t_i\}) &= \min_{0 \leq i \leq M-1} \left[t_i \cdot \log_2 \frac{1}{\sqrt{\rho_i} + \eta_i} \right] - \log_2(M+2) \\ &\geq \lambda \quad \text{by construction of } t_i \end{aligned} \quad (4)$$

$$\kappa_{\text{bind}}(n, q) = n - 3 \log_2 q - \log_2((M+2) c_b) \approx n - 3 \log_2 q - 3.3 \quad (5)$$

$$\kappa_{\text{FS}}(e, q) = 64e - 2 \log_2 K - 2 \log_2 q \approx 64e - 6.6 - 2 \log_2 q \quad (6)$$

with $\kappa_{\text{sys}} = \min(\kappa_{\text{IT}}, \kappa_{\text{bind}}, \kappa_{\text{FS}})$. The rising per-round bit yield (Theorem 4) lets each t_i shrink, so $\sum_i t_i$ is $\sim 37\%$ below FRI's uniform $r \cdot M$ (Table 1), composing with the FS-penalty improvement ($K: 18 \rightarrow 10$ at $k=4$).

4.2 Fiat–Shamir Ceiling Dictates Extension Degree

Evaluating $\kappa_{\text{FS}} = 64e - 6.6 - 2 \log_2 q$ (eq. (6)): $\mathbb{F}_{p^3}$ (2^{192}) falls below Level 1 for $q \geq 2^{40}$ ($\kappa_{\text{FS}} = 105/55/5$ at $q = 2^{40}/2^{65}/2^{90}$); $\mathbb{F}_{p^6}$ ($\kappa_{\text{FS}} = 297/247/197$) suffices for Levels 1–3, and $\mathbb{F}_{p^8}$ ($425/375/325$) is required for Level 5.

4.3 QROM Soundness

Theorem 2 (QROM soundness). *The DEEP-ALI-STIR STARK over $\mathbb{F}_{p^e}$ with folding factor $k = 4$, $M = 8$ rounds ($K = 10$), rate $\rho_0 = 1/32$, effective collision entropy n , per-query detection $\sigma_0 \geq 1 - \sqrt{\rho_0} - \eta_0 \approx 0.814$, r queries and quantum budget q achieves*

$$\kappa_{\text{sys}} = \min(2.43 r, \quad n - 3 \log_2 q - 3.3, \quad 64e - 6.6 - 2 \log_2 q).$$

Proof sketch. Instantiate Theorem 1 with the unconditional Johnson floor $\sigma_0 \geq 1 - \sqrt{\rho_0} - \eta_0$ from Theorem 4. Converting each term in (3) to bit-security yields the three ceilings (4)–(6). $\square$

4.4 Scope of the QROM Argument

Theorem 1 re-instantiates [19, Theorem 4] under BCS [3] compilation with STIR for FRI; the changes are confined to ε_{IT} (absorbing Theorem 4) and a one-bit refinement of $\varepsilon_{\text{bind}}$ ($K=10$ vs 18). The five-hybrid argument is [19, Theorem 4] unchanged (Proposition 1, Appendix A): the Zhandry collision [5] and DFMS reprogramming [6] bounds hold against arbitrary q -bounded strategies, so union gives the additive form and STIR enters only via ε_{IT} . Whether the ≈ 6.6 -bit DFMS penalty can be absorbed remains open, as for FRI [19].

Table 1. NIST-compliant parameter sets for DEEP-ALI-STIR under the unconditional Johnson bound (Theorem 4, $k=4$, $M=8$). t_0 is the worst-round query count, $\sum t_i$ the total budget; $\kappa_{\text{sys}} = \lfloor -\log_2(2^{-\kappa_{\text{IT}}} + 2^{-\kappa_{\text{bind}}} + 2^{-\kappa_{\text{FS}}}) \rfloor$ with $\kappa_{\text{IT}} = \lambda+1$. *These are the $\div 2$ declining- $\{t_i\}$ parameters; the benchmarked $\div 4$ uniform- r fold (Tables 3, 5) has $\kappa_{\text{sys}} = \min(\kappa_{\text{IT}}, \kappa_{\text{bind}}, \kappa_{\text{FS}}) \approx 130/193/259$ at $q=2^{40}$ (η -included 2.43-yield, IT-limited at every level). $\dagger$: binding wall violated.*

Level	q	Ext. Hash	t_0	$\sum t_i$	κ_{IT}	κ_{bind}	κ_{FS}	κ_{sys}
1	2^{40}	$\mathbb{F}_{p^6}$ SHA3-256	55	279	129	133	297	128
3	2^{40}	$\mathbb{F}_{p^6}$ SHA3-384	81	413	193	261	297	192
5	2^{40}	$\mathbb{F}_{p^8}$ SHA3-512	108	544	257	389	425	256
5	2^{90}	$\mathbb{F}_{p^8}$ SHA3-512	108	544	257	239	325	238[†]

5 Multi-Level Parameterisation

Combining (4)–(6) yields parameter triples (e, n, r) for each NIST level under the unconditional Johnson bound (Theorem 4).

Table 1 meets the NIST target at every non-wall-breached row. The $\div 2$ halving meets each subsequent round’s soundness target with strictly fewer queries (the declining $\{t_i\}$ of Theorem 4, summing to 279/413/544 at L1/L3/L5), so the proven budget is $\approx 37\%$ below FRI’s uniform $r \cdot M = 440/648/864$ at $M=8$ — independent of, and composing with, the constant single-leaf opening (Theorem 5) and the smaller $M=8$ at $k=4$.

Security margin: the $\div 2$ schedule (this table) vs the deployed $\div 4$.

Table 1’s at-floor figures are the $\div 2$ schedule’s — not the benchmarked system’s. The $\div 2$ declining- $\{t_i\}$ schedule tunes $\kappa_{\text{IT}} = \lambda+1$ so κ_{sys} lands exactly at the NIST floor 128/192/256 — tighter than the merged FRI variant’s 131/195/262 [19]. The *deployed* $\div 4$ uniform- r config (Tables 3, 5) reuses the worst-round t_0 as flat $r \in \{55, 81, 108\}$, the minimal sound counts at $\rho_0=1/32$ under the η -included Johnson yield (2.43 bits/query, giving $\kappa_{\text{IT}} \approx 130/193/259$ at L1/L3/L5; Theorem 4). With the schedule-independent $\kappa_{\text{bind}}=133/261/389$ and $\kappa_{\text{FS}}=297/297/425$, $\kappa_{\text{sys}} = \min(\kappa_{\text{IT}}, \kappa_{\text{bind}}, \kappa_{\text{FS}}) \approx 130/193/259$ at $q = 2^{40}$ — *IT-limited at every level*, $\lambda+2 / \lambda+1 / \lambda+3$ over the floor. (An idealized $\eta=0$ yield overstates these by $\sim \{4, 6, 8\}$ bits; the η -included count supersedes it.) The $\div 2$ schedule instead tunes κ_{IT} to exactly $\lambda+1$ for the $\approx 37\%$ query reduction.

STIR-App. C.1’s Johnson slack $\eta_i \leq \sqrt{\rho_i}/20$ is satisfied at every level (the field-size condition holds with ≈ 21 -bit margin at the tightest, L3). Per Table 1, L1/L3 hold to $q=2^{90}$ on $\mathbb{F}_{p^6}$ and L5 needs $\mathbb{F}_{p^8}$ /SHA3-512, the wall reached at $q \approx 2^{84}$ and breached at $q=2^{90}$ ($\kappa_{\text{sys}}=238$).

6 Binding Wall, FIPS-202 Limit, and Native SHA-3

Binding wall and FIPS-202 capacity limit. Inherited from [19, §5.2–5.3] (Proposition 2) are the binding-wall theorem $q_{\text{max}} = 2^{\lfloor (n-\lambda-\lceil \log_2((M+2)c_b) \rceil)/3 \rfloor}$, the sponge bound $O(q^3/2^{n_{\text{eff}}})$ with $n_{\text{eff}} = \min(n_{\text{out}}, c) \leq 512$, and the FIPS-202

limit. STIR’s smaller $K = M + 2 = 10$ (vs FRI’s 18) yields a one-bit refinement at every cell of the wall:

Table 2. $q_{\max}$ before the binding wall is breached (STIR, $M = 8$): one-bit improvement over the FRI variant ($L = 16$) [19] at every entry where the $\lfloor \cdot/3 \rfloor$ floor crosses (the SHA3-256/L1 cell ties at 2^{41}).

Hash	L1 ($\lambda=128$)	L3 ($\lambda=192$)	L5 ($\lambda=256$)
SHA3-256	2^{41}	2^{20}	< 1
SHA3-384	2^{84}	2^{62}	2^{41}
SHA3-512	2^{126}	2^{105}	2^{84}

Native SHA-3 throughout; dual-hash further demoted. [19] already defaulted to native SHA-3 on every path, offering dual-hash (Poseidon on prover-internal hashing) only as an optional $\approx 1.5\text{--}2\times$ accelerant justified by its larger Merkle workload at $r(L+1) = 55 \times 17 = 935$ openings. The deployed $\div 4$ workload drops to $r \cdot M = 55 \times 8 = 440$ openings at L1 ($\approx 2.1\times$), making the accelerant unnecessary — native SHA-3 alone is comfortable on the hot path (≈ 2.2 ms verify at L1, Table 3); the $\div 2$ schedule trims further to $\sum t_i = 279$ but pays $\approx 1.9\times$ prover time (§8.3). The base protocol uses a single SHA-3 variant across Merkle trees, FS sponge, and roots; Poseidon is retained *only* for recursive verification, where the Poseidon-erasure lemma (Proposition 3, [19, Lemma 3]) preserves the SHA-3 QROM guarantee.

7 Soundness, Zero-Knowledge, and Extractability

Lemma 2 (Perfect HVZK of the merge/LDT layer). *The honest-verifier view of the merge LDT sub-protocol is witness-independent when $N_{\text{leak}} \leq d_0$. Here $N_{\text{leak}} \leq 2 \sum_i t_i + M + (d_M + 1) \leq 631$ (L1) / 1161 (L5), both $\ll d_0 = 2^{22} - 1$, with βR masking inside the layer-0 leaf (following [19, Lem. 4]).*

This bounds leakage *from the LDT*; the *ML-DSA composite* additionally composes this layer with F2b TraceOpenings that expose witness rows for cross-sub-AIR binding (§8.2), so the end-to-end ML-DSA system is *not* zero-knowledge (the single-AIR verifiers carry no such openings). The two statements are consistent: the merge/LDT layer is perfectly HVZK, the cross-bound composite is not.

Theorem 3 (Argument of Knowledge, classical ROM). *The FS-compiled DEEP-ALI-STIR STARK is a classical-ROM argument of knowledge with error $\varepsilon_{\text{know}} \leq \varepsilon_H(q) + q^2/2^{n_{\text{out}}+1}$ via a straight-line $O(q + |H_0|)$ extractor ([19, Thm. 9]).*

Scope. QROM *soundness* holds via Theorem 4 ◦ Theorem 1 (knowledge soundness open as for FRI before [7]); plain soundness suffices for verification-of-attestation — the use case our signature AIRs serve — but not extractability-dependent deployments.

8 Empirical Validation

We implement all configurations over Goldilocks (Arkworks [11], $\mathbb{F}_{p^6}/\mathbb{F}_{p^8}$ towers, STIR at $k=4$, $M=8$, $K=10$, native SHA-3 via `tiny_keccak`; Poseidon only in recursion) on an Apple M4 Mac mini (10 cores, 16 GB RAM, 10 Rayon threads), $T \in [2^{11}, 2^{24}]$; source/data at [20]. The suite covers synthetic AIRs (Fibonacci, PoseidonChain, RegisterMachine, CairoSimple) and the four real cryptographic AIRs of §8.2, all merged-construction (eq. (2)) measurements, medians of three.

Reading guide. We keep two STIR/FRI advantages separate: mechanism (a), the analytic $\approx 37\%$ schedule-shrinkage query reduction ($\sum t_i = 279$ vs $r \cdot M = 440$ at L1, Theorem 4), and mechanism (b), the $\approx 2.1\times$ arity-port opening reduction ($k=2 \rightarrow 4$, $r(L+1) = 935 \rightarrow rM = 440$). Both commit one w -wide layer-0 leaf (no packing asymmetry, Lemma 3); (b) is equally available to FRI, while (a) is STIR’s rate-dependent LDT-byte advantage ($1.15\text{--}1.41\times$ fewer LDT bytes; Table 4).

The four LDT configurations referenced across the benchmarks are mapped explicitly in Table 6 (Appendix A).

8.1 Proof Sizes and Verification

Table 3 reports the deployed STIR configuration ($k=4$, $\div 4$, $M=8$, uniform- r); the arity-matched comparison is Table 4. The artifact also implements the $\div 2$ halving variants realising the proven $\{t_i\}$ schedule (M1): at matched arity opening a rate-dependent $1.15\text{--}1.41\times$ fewer LDT bytes than sound flat FRI- $k=4$ (an *LDT-component* ratio; at sound parameters it carries to whole-proof only marginally — $\approx 1\%$, second-order; §8.3), at $\approx 1.9\times$ prover time (full accounting in §8.3).

Table 3. Synthetic-AIR (Fibonacci) deployed-LDT proof size and timing under sound FRI- $k=4$. $T=2^{16}$, $\rho_0=1/32$, $|H_0|=2^{21}$, $\div 4$ fold; medians of three. Synthetic points use $r \in \{54, 79\}$ (illustrative width-sweep, just below the L1/L3 sound floor); the cryptographic-AIR deployment (Table 5) uses the minimum sound counts $r \in \{55, 81, 108\}$. L5 ($\mathbb{F}_{p^8}$) is reported via the ML-DSA-87 row of Table 5.

Lvl	Ext	Hash	r	Proof (KiB)	t_v (ms)	t_p (s)
1	$\mathbb{F}_{p^6}$	SHA3-256	54	587	2.17	7.4
3	$\mathbb{F}_{p^6}$	SHA3-384	79	1227	4.15	7.2

Synthetic FRI- $k=4$ proof sizes (illustrative, $r \in \{54, 79\}$ just below the L1/L3 sound floor) are 587/1,227 KiB at L1/L3 (Table 3); the deployed AIRs’ full witness-binding sound proofs (Table 5) are 9.8–325 MiB (classical) and 17–55 MiB (ML-DSA), $O(rw)$ -opening-dominated over a 0.40–2.2 MiB LDT; verify 25–930 ms, prove 0.6 s to minutes. The matched-arity comparison (Table 4) shows the same qualitative rate-dependence as [17, Table 2]: our measured FRI/STIR size ratios

Table 4. Matched-arity rate sweep: STIR vs. a sound FRI- $k=4$ baseline, both at $k=4$ through the same $\div 2$ -halving merge family; $d=2^{18}$, $\lambda=128$ (Johnson, L1/SHA3-256). These are LDT-**component** ratios (proof bytes attributable to the low-degree test) at the reduced $d=2^{18}$ proxy, *not* whole-proof figures: measured end-to-end on the real deployed AIRs, $\div 2$ ’s component advantage is second-order at whole-proof ($\approx 1\%$, §8.3). Each LDT uses its own sound schedule (STIR declining, FRI *flat* — we do *not* give FRI STIR’s schedule as an earlier baseline wrongly did). Both verify; Σt total openings, ratios FRI/STIR. STIR is smaller and faster to verify, the advantage *grows with* ρ_0 (the same qualitative trend as [17, Table 2]), at $\approx 1.9\times$ prover time (offline, amortized). High- ρ rows are an ablation; we deploy $\rho_0=1/32$.

ρ_0	STIR ($\div 2$)			FRI- $k4$		FRI/STIR		
	Σt	Proof		Σt	Proof	size	t_v	t_p
1/4	410	610 KiB		768	862 KiB	$1.41\times$	$1.45\times$	$0.54\times$
1/8	314	487 KiB		516	623 KiB	$1.28\times$	$1.31\times$	$0.53\times$
1/16	257	416 KiB		384	495 KiB	$1.19\times$	$1.23\times$	$0.53\times$
$1/32^\dagger$	219	371 KiB		312	429 KiB	$1.15\times$	$1.17\times$	$0.52\times$

† our deployed Johnson rate. Component (LDT) proof bytes only; whole-proof sizes are in Table 5. t_p ratio < 1 means FRI proves faster (STIR pays $\approx 1.9\times$).

are $1.41/1.28/1.19\times$ at $\rho=1/4/1/8/1/16$ ($d=2^{18}$). These are LDT-*component* ratios; at whole-proof, measured on the real AIRs, $\div 2$ yields only a $\approx 1\%$ (second-order) reduction (§8.3), so we deploy $\div 4$ uniform- r for its simpler soundness throughout.

Consistency and soundness regime. An earlier 206 KiB FRI figure undercounted a layer leaf; the full soundness leaf gives ≈ 740 KiB at $|H_0|=2^{21}$ ($k=2$, $L=16$) — the scale of [19]’s 1,081 KiB — and the deployed $k=4$ fold’s $\approx 2.1\times$ fewer openings carry $740 \rightarrow 587$ KiB. Both use the unconditional Johnson LDT (BCIKS Thm. 8.3 [4], not the capacity conjecture); we refuse the capacity conjecture [4, Conj. 8.4].

8.2 Real Cryptographic AIRs

The reference implementation benchmarks four *witness-binding* cryptographic verify circuits end-to-end: each AIR’s merged composition polynomial (eq. (2)) is low-degree-tested while its trace LDE is Merkle-committed and re-checked at every LDT query (the `sub_air_with_trace` path of Lemma 1), so a tampered signature makes the discarded quotient remainder non-zero and verification *rejects* — confirmed honest-accept/tamper-reject (and, for ECDSA, cross-signature reject) on every row. The LDT runs the shipped binary ($\div 2$) schedule over the merged poly at $r \in \{55, 81, 108\}$; the $\div 2$ -vs- $\div 4$ choice moves the LDT bytes by the documented $\approx 1\text{--}2\%$ (§8.3). The four AIRs: **RSA-2048** (compact exp-chain $s^{65537} \bmod n$); **ECDSA-P256** (double-scalar verify as $K=256$ double-and-add rows, public-input-bound); **Ed25519** (EdDSA with the verdict enforced at the full $k=256$ -bit scalar); and the **ML-DSA composite** (FIPS-204 §3 Algorithm 3 in-circuit, four *witness-sharing* sub-AIRs V17 (8192×323), INTT (K instances, $\sim 2048\times 260$), COEFF (three sub-AIRs, $1024\times \leq 29$), TRANSCRIPT (256×11520)).

— 9 FRI sub-proofs in all, the wide TRANSCRIPT dominating the proof via its trace openings (§8.2) — bound via the three-mechanism F2b/OOD/PI-hash composition of Contribution (2); at L1 the scheme is ML-DSA- $\{44, 65, 87\}$.

Table 5. Sound witness-binding signature-AIR proof sizes (Apple M4 Mac mini; $r \in \{55, 81, 108\}$, $\mathbb{F}_{p^6}/\mathbb{F}_{p^6}/\mathbb{F}_{p^8}$ at L1/L3/L5; honest-accept and tamper-reject on every row). π_{full} is the *complete* serialized sound proof (= LDT + $2rw$ trace-cell openings + paths), measured to the byte and *dominated by the witness openings* (the LDT is only 0.4–1.5 MiB). “calc” (✓): the single-AIR closed form (§8.4) reproduces the measured π_{full} to $< 1\%$; ML-DSA (—) is the directly-measured 9-sub-AIR composite, outside the single-AIR form. “pub”: the public input the verifier also needs (signature+key+digest). π_{full} is blowup-independent (the $O(rw)$ openings set it); t_p at a feasible blowup ($t_p \propto |H_0|$, π_{full} unchanged); ML-DSA at the deployed $b=32$. § Ed25519, ¶ ECDSA-P256 at full $K=256$; * L5 ML-DSA-87 at $\mathbb{F}_{p^8}$.

Sound AIR	meas. π_{full} (L1/L3/L5)	calc $\pi < 1\%$ (L1/L3/L5)	pub	t_v (ms)	t_p (s)
RSA-2048 exp	9.8/14.8/20.2 MiB	9.8/14.8/20.2 ✓	544 B	25/41/71	0.60/0.64/0.69
Ed25519 [§]	35.4/52.5/70.5 MiB	35.4/52.5/70.6 ✓	96 B	71/134/217	7.1/7.2/7.2
ECDSA-P256 [¶]	165/243/324 MiB	165/244/325 ✓	160 B	320/542/932	31.6/32.1/34.1
ML-DSA- $\{44, 65, 87\}$ *	17.4/31.9/55.1 MiB	— —	3.7/5.3/7.2 KB	84/149/350	24.7/28.7/57.5

Soundness is not free. The earlier non-binding merge folded the whole width- w trace into f_0 , a pure-LDT proof (≤ 445 KiB) — but it let a tampered signature verify. Re-checking the trace at each LDT query makes every classical AIR sound at the cost of a $2rw$ -cell binding payload (plus h -scaled paths) that *dominates* the full sound proof — RSA-2048 9.8, Ed25519 35, ECDSA-P256 165 MiB at L1 (validated to the byte against the serialized proof), the same trace-opening regime as the composite (§8.4). The LDT being the minor term, $\div 2$ -vs- $\div 4$ moves the full proof $< 1\%$; we report $\div 2$. Lowering the blowup re-inflates the LDT and, for the composite, the query count: at $b=4$ the v2 inner runs $r=130$ to hold L1, growing its proof to 41.8/77.0/127.7 MiB ($\approx 2.4\times$ the $b=32$, $r=54$ deployment) while proving faster on the smaller domain.

Harness notes. Tables report uniform- r folding at $r \in \{55, 81, 108\}$. The * L5 ML-DSA-87 row is at the deployed-correct $\mathbb{F}_{p^8}$; the § Ed25519 and ¶ ECDSA-P256 rows are at full $K=256$ (the ¶ double-chain multi-row AIR lays u_1G and u_2Q as $K=256$ double-and-add rows each, rather than the single-row AIR whose $\sim 10^6$ -column flat row OOMs the builder). Each π_{full} is byte-validated against the serialized proof; verify and prove times are measured medians of three at a feasible blowup (the $b=32$ ECDSA/Ed25519 builds need $\approx 25/11$ GB, but π_{full} is blowup-independent and $t_p \propto |H_0|$).

Zero-knowledge status. The ML-DSA construction is sound and classical-ROM knowledge-sound (Theorem 3; QROM open) but *not* zero-knowledge — the F2b TraceOpening exposes witness rows — which suffices for the present

public-witness use cases (DNSSEC RRSIG, rollup aggregation); private-signature flows need a separate ZK construction (not developed here).

8.3 $\div 4$ vs the $\div 2$ Proven Schedule: the LDT Is Second-Order

The $\div 2$ proven schedule opens $1.15\text{--}1.41\times$ fewer LDT leaves than flat $\div 4$ FRI- $k=4$ (Theorem 4), but the whole-proof effect is marginal even on the LDT alone ($|H_0|=2^{17}$: 445.2 vs 439.8 KiB, a $\approx 1.2\%$ $\div 2$ edge at $\approx 2\times$ slower verify) — and since the LDT is a small fraction of the trace-opening-dominated sound proof, the choice moves the total $< 1\%$. We deploy $\div 4$ FRI- $k=4$ for its *simpler unconditional soundness* (BCIKS Thm. 8.3 [4]), not a size advantage.

8.4 A Cost Model for DEEP-ALI-Merge STARKs

That the LDT choice is second-order (§8.3) points at what *does* determine cost. Because the merge collapses the width- w trace into a single composition polynomial $Q = \Phi/Z_H$ of degree $d_c(T-1)$ —a Fiat–Shamir random linear combination of the constraints (§3.1), *not* a packing—a single low-degree test certifies the whole AIR over the LDE domain $|H_0|=Tb$. This yields a cost model that predicts, for an AIR in this class, the proof size, prover time, and verification time of its NIST Level 1/3/5 STARK proof; the four signature verifiers are worked examples.

Inputs, accuracy, determinants. The model takes $(T, w, d_c, \#\text{constraints})$ and $(s, \text{ext}, \text{hash})$. The *LDT component* S_{LDT} is a closed form in $(|H_0|=Tb, s)$ with only primitive byte constants and *no fitted parameters* (eq. (7), App. B), *independent of width w* : it reproduces the measured LDT of RSA/ECDSA/Ed25519 and the $8\times$ width sweep to the byte, out-of-sample. Verify time is dominated by the same $O(rw)$ opening re-check and is therefore width-dependent (ECDSA 320 vs RSA 25 ms). The *full sound* size adds the w -dependent binding tax S_{bind} above, so width-independence of the *whole* proof holds only in the LDT-dominated regime (the synthetic sweep, $w \leq 16$); the signature AIRs are binding-dominated. *Prover time* is $\text{FFT}(|H_0|) + |H_0|c_{\text{AIR}}$ with one calibrated per-row term c_{AIR} (microbenchmark or leave-one-out): width is *relocated to constraint-evaluation cost, not removed* — FFT-bound for cheap constraints, constraint-bound for expensive ones (Ed25519 proves $\approx 6.7\times$ slower than RSA-2048 despite a $4\times$ shorter trace).

Two components: a width-independent LDT plus a width-dependent binding tax. The merge’s contribution is genuine and width-independent: it folds the width- w trace into one composition column f_0 , so a *single* low-degree test of size $S_{\text{LDT}}(|H_0|, r, M, h, E)$ (eq. (7)) certifies the whole AIR regardless of w . But making the AIR *sound* costs a second, w -dependent term: binding the witness re-opens the width- w trace row at every query, an $S_{\text{bind}} = 2rwF + (h\text{-scaled Merkle paths}) = O(rw)$ payload (validated to the byte against the serialized proof). The sound size $S_{\text{sound}} = S_{\text{LDT}} + S_{\text{bind}}$ is therefore *not* a single universal closed form; it is signature-dependent through each AIR’s width, and which term dominates flips with w : the synthetic sweep ($w \leq 16$) is LDT-dominated

and width-independent, but every real signature AIR ($w=11\text{k}–196\text{k}$) is binding-dominated — RSA-exp ($w=11,041$) 9.8, Ed25519 (41,519) 35, ECDSA-P256 (196,193) 165 MiB sound at L1, and the ML-DSA composite (17.4–55.1 MiB, $\approx 73\%$ openings, led by its SHAKE-256 TRANSCRIPT, the wide-AIR case of the same rule). The lever for smaller sound proofs is thus a *narrower* AIR (bit-packing, fewer columns), not a different low-degree test. We provide a Python implementation (`sound_proof_calc.py` in [20]) that computes S_{sound} for any AIR from its (T, b, w, r) and the level parameters (h, E) , self-validating against the byte anchors and reproducing every measured single-AIR proof to $<1\%$ (the *calc* column of Table 5).

Width sweep. Three AIRs across an $8\times$ width range at fixed $|H_0|=2^{22}$ ($r=54$, $b=32$, $\mathbb{F}_{p^8}$, $k=4$, $M=\log_2(|H_0|/|H_M|)=11$ rounds; a high-blowup/low- r point chosen to isolate width, so its absolute figures are config-specific, not the deployed tables’) — Fibonacci ($w=2$, 1 constraint), RegisterMachine ($w=8$, 8), PoseidonChain ($w=16$, 16) — give a *byte-identical* 373,152-byte proof, verify within 1.2% (1.43–1.45 ms) and prove within 0.3% (10.04–10.06 s). Width enters neither size nor verify, and enters prover only through c_{AIR} —negligible here, so prover $\approx \text{FFT}(|H_0|)$. This is a controlled sweep at a single constant-rate $k=4$ configuration; the invariant is byte-identity *across width*, not the absolute figures, which are configuration-specific and not comparable with the deployed Tables 3 and 5 at their own $(r, \text{ext}, |H_0|)$.

Reducing the binding payload (future work). As $S_{\text{bind}}=2rwF$ dominates the sound proof, every lever is width-side: a *narrower* verifier circuit (bit-packed Keccak state, fewer auxiliary columns); a column-wise (vector) trace commitment that opens only the constraint-referenced columns instead of the full width- w row; dropping the next-row opening for *local*-constraint AIRs (a $\frac{1}{2}\times$ saving, available now); and batched Merkle openings sharing path nodes. The merge already makes the LDT minimal; a *width-efficient witness binding* is the open problem these proofs pose.

Threats to validity. Proof size is a hardware-independent byte count (the closed form is exact for the LDT component on any platform; the sound size adds the measured $2rwF$); the prover and verification *times* are wall-clock and depend on the hardware and Rayon thread count (we measure on an Apple M4 Mac mini, 10 threads), so the time model is calibrated per platform and predicts scaling and structure, not platform-independent absolute times. The validation domain is signature-style and synthetic AIRs (widths 2–16, moderate constraint cost over $\mathbb{F}_p$); extrapolation to very wide, low-degree, or deeply recursive AIRs is unvalidated.

8.5 Further empirical results

Three secondary results (full tables in [20]): the byte-identical-across-width LDT invariance of §8.4 also holds at L3/L5 (prover $< 7\%$ within each level); a signature rollup wrapping $N=4$ ML-DSA-44 inners gives 789 KiB/signature at L1; and a Poseidon-committed inner recursion costs $\approx 48\text{k}$ vs $\approx 5.6\times 10^5$ SHA-3 constraints ($\approx 12\times$), the outer SHA-3 proof preserving the QROM trust chain.

8.6 Arity, Blowup, and NIST-Level Sweep

We swept arity $k \in \{2, 4, 8, 16\}$ against blowup $b \in \{4, 8, 16, 32, 64\}$ over 8 (level, hash, q) cells (CSV in [20]; configurations in Table 6). The deployed $k=4$, $\div 4$ config’s round-count advantage comes from the DFMS penalty reduction ($8.3 \rightarrow 6.6$ bits) and $K \approx 10$ -vs-18 FS tightening, both rate-schedule-independent; $k=2$ collapses to FRI and erases them.

8.7 Implementation Status

The reference implementation realises eq. (2) directly: the layer-0 leaf packs $(T_1, \dots, T_w, Q, R)$ (Lemma 3), FS challenges in $\mathbb{F}_{p^e}$, the merge over $\mathbb{F}_{p^6}$ (L1/L3) or $\mathbb{F}_{p^8}$ (L5) at $k=4$, reference $M=8$, $K=10$, $r \in \{55, 81, 108\}$ (HVZK by βR ; ML-DSA cross-binding reuses the merge OOD, Lemma 1). Three milestones landed: *M1* the $\div 2$ prover; *M2* the Ed25519/ECDSA-p256 $K=256$ harnesses; *M3* $\mathbb{F}_{p^8}$ at L5 (per-AIR caveats in Table 5).

9 Related Work

No prior system is simultaneously FIPS-only on both paths, constant in authentication-path count, multi-level post-quantum at an unconditional Johnson LDT, and equipped with a formal QROM analysis (full trust-chain comparison in [20]). STIR [17] cuts query counts via out-of-domain corrections at list-decoding capacity, whose proven constants hold only for small arities — so we invoke only the Johnson bound, revising [19]’s “complementary bottleneck” framing: the gain is a $1.15\text{--}1.41 \times$ LDT-*component* reduction that does not reach whole-proof (§8.3). STARKs/FRI [1,2] build on IOPs [3]; ethSTARK [16], Plonky2/3 [12,13] and STWO [10] use algebraic hashes (Plonky3’s Keccak mode gives no formal PQ/binding-wall analysis and, without the merge, needs $\sum_i k_i$ paths per query); Binius [18] has no published QROM bound. Our analysis builds on Zhandry’s compressed oracle [5] and DFMS [6]. Unlike operation-level cost estimators (e.g. zkalc) and the ethSTARK/FRI/STIR parameter calculators, which size a *fixed* protocol from its parameters, our model is *AIR-keyed*: a closed-form prediction of LDT proof size and verify time from the trace length (width-independent for the LDT) across NIST Levels 1/3/5.

10 Conclusion

We instantiated four *full* signature AIRs (classical RSA/ECDSA/Ed25519 and FIPS-204 ML-DSA) in a FIPS-aligned STIR merge at $k=4$, validating a *cost model*: the merge makes LDT proof size and verify time width-independent closed forms in the trace length ($8 \times$ sweep byte-identical), prover time a function of constraint complexity; the ML-DSA *composite* is instead trace-opening-dominated — its cross-sub-AIR binding opens trace rows, inherent to a multi-sub-AIR composite, so a narrower transcript AIR (not a different LDT) is the lever for smaller proofs. We deploy the unconditionally sound $\div 4$, refuse the capacity conjecture [4, Conj. 8.4], and leave QROM knowledge soundness and ZK open.

A Inherited results (companion [19])

We state, as propositions, the load-bearing results inherited unchanged from the companion [19] so the security argument is checkable without it; full proofs are in that companion.

Proposition 1 (Three-pillar QROM decomposition). *For the FS-compiled merge STARK over $\mathbb{F}_{p^e}$ with hash H of effective collision entropy $n = \min(n_{\text{out}}, c)$ and any (q, t) -QROM adversary, $\varepsilon_{\Pi}(q) \leq \varepsilon_{\text{IT}} + \varepsilon_{\text{bind}} + \varepsilon_{\text{FS}}$ with $\varepsilon_{\text{bind}} = (M+2)c_b q^3/2^n$ and $\varepsilon_{\text{FS}} = K^2(q+1)^2/|\mathbb{F}_{p^e}|$ (eq. (3)). The bound follows a five-hybrid argument (ideal Merkle, DEEP conditioning, IT soundness, DFMS reprogramming, challenge re-injection) whose collision [5] and reprogramming [6] steps hold against arbitrary q -bounded strategies, so a union over the transcript-event partition is additive. (This is [19, Theorem 4] with FRI replaced by STIR; only ε_{IT} changes, absorbing Theorem 4.)*

Proposition 2 (Binding-wall theorem). *The largest quantum budget before the binding term breaches the NIST floor is $q_{\text{max}} = 2^{\lfloor (n-\lambda - \lceil \log_2((M+2)c_b) \rceil)/3 \rfloor}$, from the sponge bound $\varepsilon_{\text{col}}^{\text{pq}} \leq O(q^3/2^{n_{\text{eff}}})$ with $n_{\text{eff}} = \min(n_{\text{out}}, c) \leq 512$ across all FIPS-202 functions (Table 2). (Invert $\kappa_{\text{bind}} \geq \lambda$, eq. (5); [19, §5.2–5.3].)*

Proposition 3 (Poseidon-erasure lemma). *A recursive verifier may commit its inner transcript under Poseidon provided the outer proof re-commits under SHA-3: the inner Poseidon root is erased from the QROM trust chain at the outer boundary, so the post-quantum guarantee rests on SHA-3 collision resistance alone. (The outer SHA-3 commitment is binding in the QROM [5] independently of the inner hash; [19, Lem. 3].)*

Proposition 4 (DEEP-ALI degree bound and initial distance). *For an AIR with s constraints of degree $\leq d_c$ over the trace subgroup H of order T (merge target $d_0 = (d_c-1)T - 1$), let $\varepsilon_{\text{ali}} = (s + T + 1)/(|\mathbb{F}_{p^e}| - |H_0|)$ (the combination contributes $s/|\mathbb{F}_{p^e}|$; the OOD check evades with probability $\leq T/(|\mathbb{F}_{p^e}| - |H_0|)$ since the remainder $R = \Phi \bmod Z_H$ has $\deg R \leq T-1$). Then (i) a valid AIR gives $f_0 \in \text{RS}_{H_0}(d_0)$ and satisfies (1), and (ii) an invalid AIR has $\text{dist}(f_0, \text{RS}_{H_0}(d_0)) \geq 1 - \rho_0 - \varepsilon_{\text{ali}}$ except with probability $\leq \varepsilon_{\text{ali}}$. (This is [19, Theorem 1] with the STIR proximity test for FRI and target degree the ALI-quotient bound; the case split on combination cancellation ($\leq s/|\mathbb{F}_{p^e}|$) and consistency-check evasion ($\leq T/(|\mathbb{F}_{p^e}| - |H_0|)$ by $\deg R \leq T-1$) is unchanged.)*

B STIR-instantiation soundness (this work)

The merge’s STIR-instantiation results summarised in §3.3; the distance and single-leaf facts adapt the companion’s binary statements to arity k , the contraction is our re-derivation.

Table 6. Configuration map across the four LDT configurations referenced in this paper. Entries at L1, $n_0 = 2^{17}$, $\rho_0 = 1/32$. The “ $\div 4$ uniform” row is deployed (Tables 3, 5); “ $\div 2$ secured” is the rate-improving schedule of §8.3 (a $1.15\text{--}1.41\times$ LDT-component ratio that attenuates to $\approx 1\%$ at whole-proof). All rows share $M=8$ (set by the domain ratio $\log_2(|H_0|/|H_M|)$, not the per-round rate), so the binding-wall, DFMS, and FS-ceiling improvements hold in either schedule.

Config	k	dom $\div$ /rd	M rate schedule	query schedule
$\div 4$ uniform (deployed, Tables 3, 5)	4	4	8 constant ρ_0	$r=55$ uniform
$\div 2$ secured (implemented, not deployed, §8.3)	4	2	$8 \rho_0 \cdot 2^{-i}$	$\{55, 46, 39, 34, 30, 27, 25, 23\}$
$k=2$ uniform STIR (§8.6)	2	2	17 constant ρ_0	$r=55$ uniform
FRI uniform (matched-Johnson)	2	2	17 constant ρ_0	$r=55$ uniform

Theorem 4 (STIR Johnson-regime per-round contraction). *Under STIR’s $\div 2$ domain-halving with folding factor k , the round- i rate is $\rho_i = \rho_0(2/k)^i$; under the unconditional Johnson-regime proximity gap at rate ρ_i the per-query rejection is $\sigma_i \geq 1 - \sqrt{\rho_i} - \eta_i^J$ with $\eta_i^J \leq \sqrt{\rho_i}/20$ [17, App. C.1], giving the schedule $t_i = \lceil (\lambda + \log_2(M+2) + 1) / (-\log_2(\sqrt{\rho_i} + \eta_i^J)) \rceil$ (Table 1). The numerator carries an extra $\log_2(M+2)$ -bit margin over STIR’s recommended $\lambda + 1$ [17, §5.3], accommodating the bundled IT bound of Theorem 1.*

Proof (induction step; our derivation). Base ($i=0$): at rate ρ_0 the Johnson per-query yield is $-\log_2(\sqrt{\rho_0} + \eta_0^J)$, so t_0 queries certify $t_0(-\log_2(\sqrt{\rho_0} + \eta_0^J)) \geq \lambda + \log_2(M+2) + 1$ bits. Step ($i \rightarrow i+1$): the round- i fold divides degree by k and the domain by 2, so $\rho_{i+1} = \rho_i(2/k) < \rho_i$; by STIR’s round-consistency [17, Thm. 5.1] a word δ_i -far from the round- i code folds to one at least Johnson(ρ_{i+1})-far except with the proximity-gap error, and since $\rho_{i+1} < \rho_i$ the floor $1 - \sqrt{\rho_{i+1}}$ rises — the per-query yield grows, so t_{i+1} from the displayed formula again certifies $\geq \lambda + \log_2(M+2) + 1$ bits. Union: over the M rounds plus the terminal polynomial, the $\log_2(M+2)$ numerator margin keeps the total IT error $\leq 2^{-(\lambda+1)} = \varepsilon_{IT}$ (Theorem 1); field-error slack ($\approx 2^{-359}$ at L1) absorbs into ε_{IT} . $\square$

Theorem 5 (Total Merkle openings: declining vs. flat). *Both LDTs open one path per query per round (Lemma 3) and query every round (the per-round IT term of Theorem 1 forbids omission), so Opens = $\sum_i t_i$: FRI’s flat t_{FRI} gives $t_{FRI}M$, STIR’s declining $\{t_i\}$ gives $\sum_i t_i \in \{279, 413, 544\}$ versus FRI’s $\{440, 648, 864\}$, and the deployed $\div 4$ holds the rate constant (structurally FRI- $k=4$). Every $t_i > 0$ — this is STIR’s mechanism (later rounds test a lower-rate, larger-distance code), not a removal of openings; the ratio grows with ρ_0 as in [17, Table 2].*

Lemma 3 (Single Merkle opening per round- i query). *At round i , query position $x_i \in H_i$ requires one opening into root_i — the k -coset leaf ($\{f_i(\zeta x_i) : \zeta^k=1\}, \text{CP}_i(x_i)$), where $\text{CP}_i(X) := \beta_i(f_i(X) - f_i(r_i)) / (X - r_i)$ is the round- i DEEP shift quotient ($\beta_i \in \mathbb{F}_{p^e}^\times$ a randomiser) — plus the next-round value $f_{i+1}(x_i^k)$ as an OOD reply. At round 0 the leaf packs the width- w row $(T_j(x_0))_{j \leq w}$ with $Q(x_0), R(x_0), f_0(x_0)$ via (2); co-committing R preserves the secrecy of Lemma 2. From that one leaf and the OOD reply the verifier checks both round- i relations*

(fold-and-correct onto H_{i+1} and the DEEP shift) without a second opening. The per-leaf cost grows with k (the leaf holds k values plus CP_i): at $k=4$ it is $\approx 2\times$ larger than at binary arity, offsetting the path-count reduction.

Lemma 4 (Composition polynomial distance). *If fold consistency is violated at any k -coset of round i , then $\text{dist}(\text{CP}_i|_{H_i}, \text{RS}_{H_i}(d_i-1)) \geq k/|H_i|$ (generalising the binary $2/|H_\ell|$ of [19, Lem. 2] to arity k); STIR’s round-by-round soundness [17, Lem. 5.4] amplifies this to the Johnson floor used in Theorem 4.*

Public-parameter binding. The Fiat–Shamir sponge absorbs the canonical $\text{pp} = (p, e, \text{H}_{\text{id}}, n_{\text{out}}, M, k, (t_i)_i, T, d_c, \rho_0)$ as the zeroth message, before root_0 ; every challenge then depends on pp , so a proof under one parameter set cannot verify under another absent a collision in H — closing a DFMS [6] requirement and separating levels (FIPS 140-3 module separation).

Proof-size closed form. For the deployed FRI- $k=4$ proof the serialization of §8.1 sums four groups:

$$S = \underbrace{h(M+2) + E(|\tau|+M+2)}_{\text{digests, roots, terminal, OOD}} + \underbrace{3ErM}_{k\text{-coset leaves}} + \underbrace{h(r(M+1) + A)}_{\text{Merkle paths}}, \quad (7)$$

with $F=8\text{ B}$ (field), $E=eF$ (48/64 B), $h \in \{32, 48, 64\}\text{ B}$ (digest); M fold rounds, r queries, $|\tau|$ terminal degree, A total path-sibling hashes (the “+2” are the config and coefficient digests and two OOD evals). Trace padding to $T=2^t$ makes every $|H_i|$ a power of two, so $M, r(M+1), A$ are exact integers fixed by $(|H_0|, k, s)$: byte-exact within a 2^n bracket. RSA-2048 ($h=32, E=48, r=55$ at L1) with $M=9, |\tau|=1, A=11,440$ for the $|H_0|=2^{17}$ reference gives $352 + 576 + 71,280 + 383,680 = 455,888\text{ B}=445.2\text{ KiB}$. The deployed witness-binding exp-chain AIR has a shorter $|H_0|=2^{10}$ trace, whose LDT the same closed form puts at the 403 KiB L1 cell of Table 5 (the witness now ships separately as the $2rw$ trace openings of §8.2). A multi-sub-AIR composite (ML-DSA: 9 FRI sub-proofs from 4 witness-sharing AIRs) cannot fold its trace into a single f_0 ; its cross-sub-AIR binding opens the width- w row per query, an $O(rw)$ term that dominates ($\approx 73\%$, bare LDT $\approx 8\%$) and breaks width-independence (§8.4).

References

1. Ben-Sasson, E., Bentov, I., Horesh, Y., Riabzev, M.: Scalable, transparent, and post-quantum secure computational integrity. ePrint 2018/046 (2018)
2. Ben-Sasson, E., Bentov, I., Horesh, Y., Riabzev, M.: Fast Reed–Solomon interactive oracle proofs of proximity. In: ICALP 2018. LIPIcs, vol. 107, pp. 14:1–14:17 (2018)
3. Ben-Sasson, E., Chiesa, A., Spooner, N.: Interactive oracle proofs. In: TCC 2016-B. LNCS, vol. 9986, pp. 31–60. Springer (2016)
4. Ben-Sasson, E., Carmon, D., Ishai, Y., Kopparty, S., Saraf, S.: Proximity gaps for Reed–Solomon codes. In: FOCS 2020. pp. 900–909. IEEE (2020). Full version: ePrint 2020/654.
5. Zhandry, M.: How to record quantum queries, and applications to quantum indistinguishability. In: CRYPTO 2019. LNCS, vol. 11693, pp. 239–268. Springer (2019)

6. Don, J., Fehr, S., Majenz, C.: The measure-and-reprogram technique 2.0: Multi-round Fiat–Shamir and more. In: CRYPTO 2020. LNCS, vol. 12172, pp. 602–631. Springer (2020)
7. Chiesa, A., Manohar, P., Spooner, N.: Succinct arguments in the quantum random oracle model. In: TCC 2019. LNCS, vol. 11892, pp. 1–29. Springer (2019)
8. Grassi, L., Khovratovich, D., Rechberger, C., Roy, A., Schofnegger, M.: Poseidon: A new hash function for zero-knowledge proof systems. In: USENIX Security 2021, pp. 519–535 (2021)
9. Bariant, A., Bouvier, C., Leurent, G., Peyrin, T.: Algebraic attacks against some arithmetization-oriented primitives. In: EUROCRYPT 2024. LNCS, vol. 14651, pp. 3–32. Springer (2024)
10. Haböck, U., Levit, D., Papini, S.: Circle STARKs. ePrint 2024/278 (2024)
11. Arkworks contributors: Arkworks: An ecosystem for zkSNARKs. <https://github.com/arkworks-rs> (2022)
12. Polygon Zero Team: Plonky2: Fast recursive arguments with PLONK and FRI. <https://polygon.technology/blog/introducing-plonky2> (2022)
13. Polygon Labs: Plonky3: A toolkit for polynomial IOPs. <https://github.com/Plonky3/Plonky3> (2024)
14. Winterfell Team: Winterfell: A STARK prover. <https://github.com/facebook/winterfell> (2021)
15. National Institute of Standards and Technology: SHA-3 Standard: Permutation-based hash and extendable-output functions. FIPS PUB 202 (2015)
16. StarkWare: ethSTARK documentation. ePrint 2021/582 (2021)
17. Arnon, G., Chiesa, A., Fenzi, G., Yogev, E.: STIR: Reed–Solomon proximity testing with fewer queries. In: CRYPTO 2024. LNCS, vol. 14929, pp. 380–413. Springer (2024). Full version: ePrint 2024/390 (cited numbers follow it).
18. Diamond, B. E., Posen, J.: Succinct arguments over towers of binary fields. In: EUROCRYPT 2025, pp. 93–122. Springer, Cham (2025)
19. Holmes, S.: FIPS-Aligned STARKs with Concrete Multi-Level Post-Quantum Security. In: Computer Security — ESORICS 2026, LNCS, Springer (2026, to appear)
20. Holmes, S.: STIR-based source code and benchmarks (2026), <https://anonymous.4open.science/r/dam-ldt-arity-bench-86C1>